



**AI-DRIVEN**

# CYBER SECURITY

*A COMPANY CAN HAVE THOUSANDS OF SECURITY ALERTS EVERY DAY—BUT A SOC MUST DECIDE WHICH ONE MEANS ‘BUSINESS AS USUAL’ AND WHICH ONE MEANS ‘WE’RE UNDER ATTACK.’*

# WHAT IS A SOC?

- SECURITY OPERATIONS CENTER
- CENTRAL SECURITY TEAM
- 24/7 MONITORING
- DETECT, INVESTIGATE & RESPOND
- PROTECTS SYSTEMS AND DATA

# SOC TEAM ROLES

- SOC ANALYST
  - TIER 1 — ALERT MONITORING
  - TIER 2 — INVESTIGATION
  - TIER 3 — ADVANCED THREAT HUNTING
- SOC MANAGER

# SECURITY LOGS

- RECORD OF SYSTEM ACTIVITY
- LOGIN EVENTS
- NETWORK ACTIVITY
- APPLICATION EVENTS
- FIREWALL EVENTS
- AUTHENTICATION EVENTS

# WHAT IS SIEM?

- SECURITY INFORMATION AND EVENT MANAGEMENT
- COLLECTS SECURITY LOGS
- CENTRALIZES DATA
- CORRELATES EVENTS
- DETECTS SUSPICIOUS ACTIVITY
- GENERATES ALERTS



# ALERT

- WARNING ABOUT SUSPICIOUS ACTIVITY
- CREATED BY DETECTION RULES
- CONTAINS SECURITY EVIDENCE
- NEEDS INVESTIGATION
- NOT EVERY ALERT IS AN ATTACK

# DETECTION

- RULES & ANALYTICS
- INDICATORS OF COMPROMISE (IOCS)
- SUSPICIOUS PATTERNS
- BEHAVIORAL ANALYSIS
- CORRELATION

# FALSE POSITIVE VS TRUE POSITIVE

- TRUE POSITIVE → REAL THREAT
- FALSE POSITIVE → NORMAL ACTIVITY DETECTED AS THREAT
- ANALYSTS VALIDATE ALERTS
- REDUCE FALSE POSITIVES



# THREAT TRIAGE

- QUICKLY ASSESS AN ALERT
- CHECK SEVERITY
- CHECK EVIDENCE
- IDENTIFY AFFECTED ASSETS
- DECIDE NEXT ACTION

# SEVERITY & PRIORITY

- LOW
- MEDIUM
- HIGH
- CRITICAL
- BUSINESS IMPACT MATTERS

# INCIDENT

- CONFIRMED SECURITY EVENT
- REQUIRES INVESTIGATION
- MAY REQUIRE CONTAINMENT
- DOCUMENTED BY SOC

# ESCALATION

- MOVE COMPLEX CASES TO HIGHER TIER
- TIER 1 → TIER 2
- TIER 2 → TIER 3
- CRITICAL INCIDENTS → INCIDENT RESPONSE TEAM
- FAST COMMUNICATION

# SOC WORKFLOW

LOG → SIEM → DETECTION → ALERT → TRIAGE →  
INVESTIGATION → ESCALATION → RESPONSE →  
RECOVERY



# INCIDENT RESPONSE

- IDENTIFY
- CONTAIN
- ERADICATE
- RECOVER
- LEARN & IMPROVE